In [1]:
```
print("Hello, World!")
```

```
Hello, World!
```

In [2]:
```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from scipy import stats
import statsmodels.api as sm
from statsmodels.formula.api import ols
from math import log, tanh
import numpy as np
```

In [3]:
```
Wage = pd.read_csv("ISLR/data/Wage.csv", index_col=0)
Wage.head()
```

Out[3]:

	year	age	maritl	race	education	region	jobclass	health	health_ins	logwage	wage
231655	2006	18	1. Never Married	1. White	1. < HS Grad	2. Middle Atlantic	1. Industrial	1. <=Good	2. No	4.318063	75.043154
86582	2004	24	1. Never Married	1. White	4. College Grad	2. Middle Atlantic	2. Information	2. >=Very Good	2. No	4.255273	70.476020
161300	2003	45	2. Married	1. White	3. Some College	2. Middle Atlantic	1. Industrial	1. <=Good	1. Yes	4.875061	130.982177
155159	2003	43	2. Married	3. Asian	4. College Grad	2. Middle Atlantic	2. Information	2. >=Very Good	1. Yes	5.041393	154.685293
11443	2005	50	4. Divorced	1. White	2. HS Grad	2. Middle Atlantic	2. Information	1. <=Good	1. Yes	4.318063	75.043154

In [4]:
```
print("Shape of Wage: ", Wage.shape[1])
print("Features of Wage: ", list(Wage.columns))
```

```
Shape of Wage:  11
Features of Wage:  ['year', 'age', 'maritl', 'race', 'education', 'region', 'jobclass', 'health', 'health_ins', 'log
wage', 'wage']
```

In [5]:
```
Wage = Wage.drop(columns=["logwage"])
print("Shape of Wage, after dropping logwage: ", Wage.shape[1])
print("Features of Wage, after dropping logwage: ", list(Wage.columns))
```

```
Shape of Wage, after dropping logwage:  10
Features of Wage, after dropping logwage:  ['year', 'age', 'maritl', 'race', 'education', 'region', 'jobclass', 'hea
lth', 'health_ins', 'wage']
```

In [6]:
```
Wage.shape[0]
```

Out[6]:
```
3000
```

In [7]:
```
Wage
```

Out[7]:

	year	age	maritl	race	education	region	jobclass	health	health_ins	wage
231655	2006	18	1. Never Married	1. White	1. < HS Grad	2. Middle Atlantic	1. Industrial	1. <=Good	2. No	75.043154
86582	2004	24	1. Never Married	1. White	4. College Grad	2. Middle Atlantic	2. Information	2. >=Very Good	2. No	70.476020
161300	2003	45	2. Married	1. White	3. Some College	2. Middle Atlantic	1. Industrial	1. <=Good	1. Yes	130.982177
155159	2003	43	2. Married	3. Asian	4. College Grad	2. Middle Atlantic	2. Information	2. >=Very Good	1. Yes	154.685293
11443	2005	50	4. Divorced	1. White	2. HS Grad	2. Middle Atlantic	2. Information	1. <=Good	1. Yes	75.043154
...	...	...	...	...	...	...	...	...	...	...
376816	2008	44	2. Married	1. White	3. Some College	2. Middle Atlantic	1. Industrial	2. >=Very Good	1. Yes	154.685293
302281	2007	30	2. Married	1. White	2. HS Grad	2. Middle Atlantic	1. Industrial	2. >=Very Good	2. No	99.689464
10033	2005	27	2. Married	2. Black	1. < HS Grad	2. Middle Atlantic	1. Industrial	1. <=Good	2. No	66.229408
14375	2005	27	1. Never Married	1. White	3. Some College	2. Middle Atlantic	1. Industrial	2. >=Very Good	1. Yes	87.981033
453557	2009	55	5. Separated	1. White	2. HS Grad	2. Middle Atlantic	1. Industrial	1. <=Good	1. Yes	90.481913

3000 rows × 10 columns

In [8]:
```
Wage.describe(include='all')
```

Out[8]:

	year	age	maritl	race	education	region	jobclass	health	health_ins	wage
count	3000.000000	3000.000000	3000	3000	3000	3000	3000	3000	3000	3000.000000
unique	NaN	NaN	5	4	5	1	2	2	2	NaN
top	NaN	NaN	2. Married	1. White	2. HS Grad	2. Middle Atlantic	1. Industrial	2. >=Very Good	1. Yes	NaN
freq	NaN	NaN	2074	2480	971	3000	1544	2142	2083	NaN
mean	2005.791000	42.414667	NaN	NaN	NaN	NaN	NaN	NaN	NaN	111.703608
std	2.026167	11.542406	NaN	NaN	NaN	NaN	NaN	NaN	NaN	41.728595
min	2003.000000	18.000000	NaN	NaN	NaN	NaN	NaN	NaN	NaN	20.085537
25%	2004.000000	33.750000	NaN	NaN	NaN	NaN	NaN	NaN	NaN	85.383940
50%	2006.000000	42.000000	NaN	NaN	NaN	NaN	NaN	NaN	NaN	104.921507
75%	2008.000000	51.000000	NaN	NaN	NaN	NaN	NaN	NaN	NaN	128.680488
max	2009.000000	80.000000	NaN	NaN	NaN	NaN	NaN	NaN	NaN	318.342430

In [9]:

```
fig, ax = plt.subplots(figsize=(10, 6))
sns.regplot(x='age', y='wage', data=Wage, scatter_kws={'alpha':0.3}, line_kws={'color':'red'}, ax=ax)
r, p = stats.pearsonr(Wage['age'], Wage['wage'])
ax.set_title(f'Age vs Wage (Pearson r = {r:.4f}, p = {p:.2e})')
ax.set_xlabel('Age')
ax.set_ylabel('Wage')
plt.tight_layout()
plt.show()
```



In [10]:
```
sample_size = min(5000, len(Wage))
age_sample = Wage['age'].sample(sample_size, random_state=42)
wage_sample = Wage['wage'].sample(sample_size, random_state=42)

stat_age, p_age = stats.shapiro(age_sample)
stat_wage, p_wage = stats.shapiro(wage_sample)
```

```
print("Shapiro-Wilk Normality Test for Age:")
print(f"  W = {stat_age:.4f}, p-value = {p_age}")
print(f"  {'Not normal' if p_age < 0.05 else 'Normal'} (at α=0.05)")

print("\nShapiro-Wilk Normality Test for Wage:")
print(f"  W = {stat_wage:.4f}, p-value = {p_wage}")
print(f"  {'Not normal' if p_wage < 0.05 else 'Normal'} (at α=0.05)")
```

```
Shapiro-Wilk Normality Test for Age:
  W = 0.9911, p-value = 9.416337654404108e-13
  Not normal (at α=0.05)

Shapiro-Wilk Normality Test for Wage:
  W = 0.8796, p-value = 2.463070103678234e-43
  Not normal (at α=0.05)
```

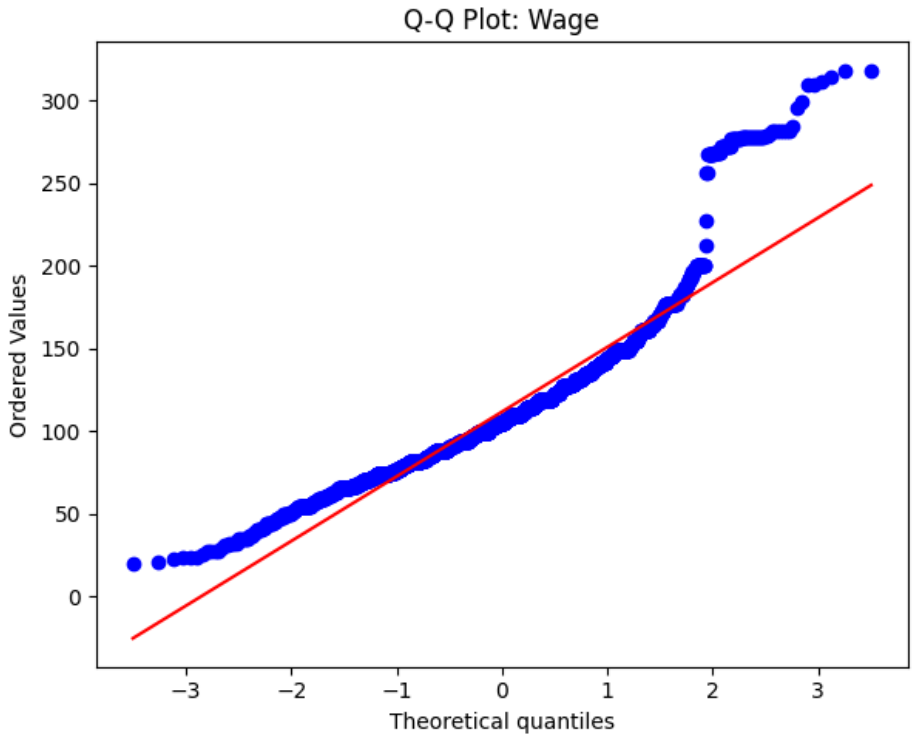
In [11]:
```
fig, axes = plt.subplots(1, 2, figsize=(12, 5))

stats.probplot(Wage['age'], dist="norm", plot=axes[0])
axes[0].set_title('Q-Q Plot: Age')

stats.probplot(Wage['wage'], dist="norm", plot=axes[1])
axes[1].set_title('Q-Q Plot: Wage')

plt.tight_layout()
plt.show()
```





In [12]:

```
r, p = stats.pearsonr(Wage['age'], Wage['wage'])
n = len(Wage)

se = 1 / ((n - 3) ** 0.5)
z = 0.5 * np.log((1 + r) / (1 - r))
z_lower = z - 1.96 * se
z_upper = z + 1.96 * se
r_lower = np.tanh(z_lower)
r_upper = np.tanh(z_upper)

print("Pearson's product-moment correlation")
print("=" * 45)
print(f"data: Wage['age'] and Wage['wage']")
print(f"t = {r * ((n-2)/(1-r**2))**0.5:.3f}, df = {n-2}, p-value = {p:.2e}")
print(f"alternative hypothesis: true correlation is not equal to 0")
print(f"95 percent confidence interval: [{r_lower:.7f}, {r_upper:.7f}]")
print(f"sample estimate (cor): {r:.7f}")
```

```
Pearson's product-moment correlation
=============================================
data: Wage['age'] and Wage['wage']
t = 10.923, df = 2998, p-value = 2.90e-27
alternative hypothesis: true correlation is not equal to 0
95 percent confidence interval: [0.1609771, 0.2298154]
sample estimate (cor): 0.1956372
```

In [13]:
```
print("Unique values for 'education':")
print(Wage['education'].unique())
```

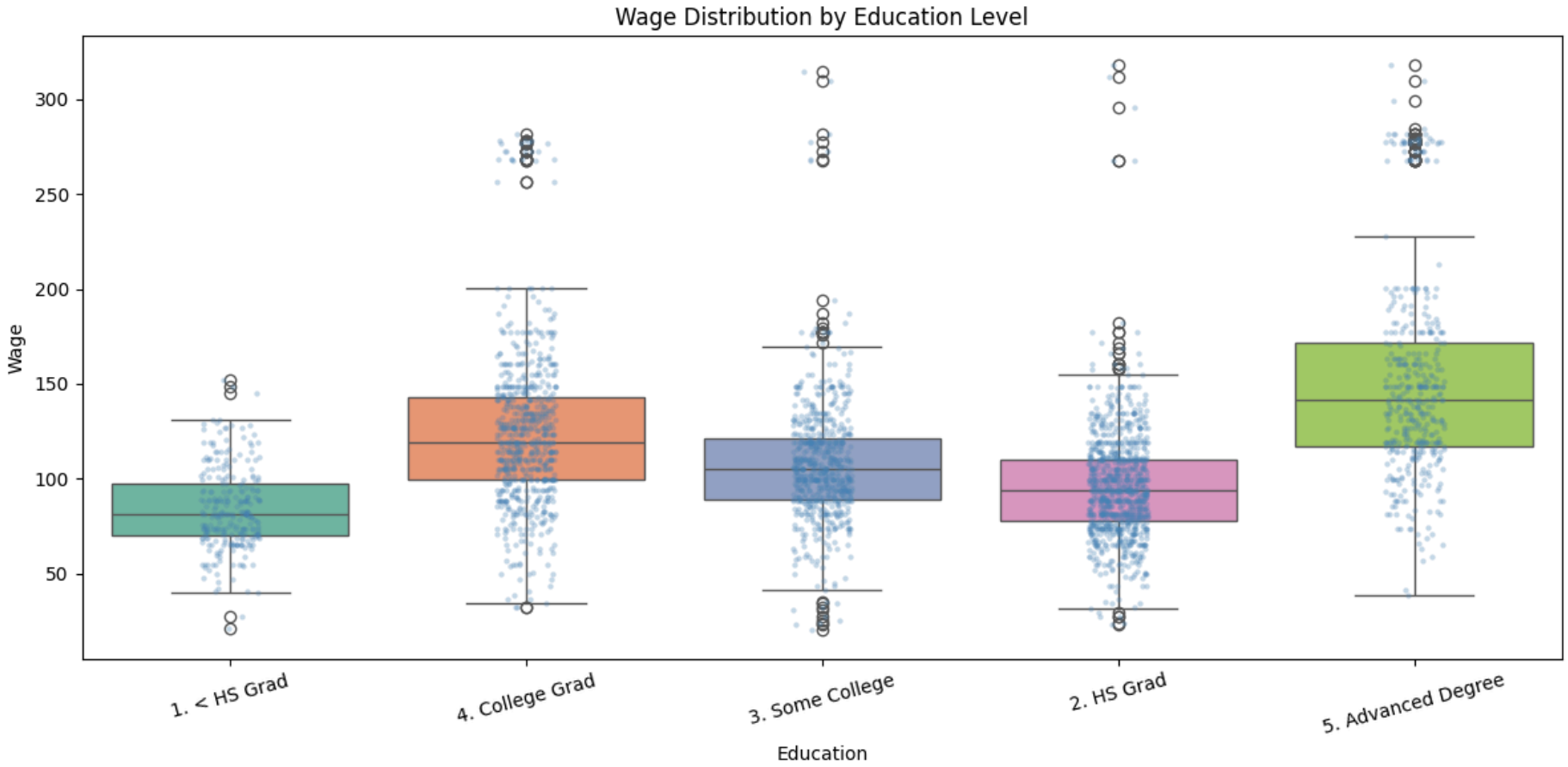
```
Unique values for 'education':
<StringArray>
[      '1. < HS Grad',    '4. College Grad',    '3. Some College',
         '2. HS Grad', '5. Advanced Degree']
Length: 5, dtype: str
```

In [14]:
```
fig, ax = plt.subplots(figsize=(12, 6))
sns.boxplot(x='education', y='wage', data=Wage, palette='Set2', ax=ax)
sns.stripplot(x='education', y='wage', data=Wage, color='steelblue', alpha=0.3, size=3, ax=ax)
ax.set_xlabel('Education')
ax.set_ylabel('Wage')
ax.set_title('Wage Distribution by Education Level')
plt.xticks(rotation=15)
plt.tight_layout()
plt.show()
```

```
/var/folders/2z/bd17dm2577gbxxvhvfvsj2lw0000gn/T/ipykernel_10893/3277231505.py:2: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `x` variable to `hue` and set `legend=False` for the same effect.

  sns.boxplot(x='education', y='wage', data=Wage, palette='Set2', ax=ax)
```



In [15]:
```
model = ols('wage ~ C(education)', data=Wage).fit()
anova_table = sm.stats.anova_lm(model, typ=2)
print("One-way ANOVA: Wage ~ Education")
print("=" * 60)
print(anova_table)
print("\nSignificance codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1")
```

```
One-way ANOVA: Wage ~ Education
============================================================
                    sum_sq      df           F         PR(>F)
C(education)  1.226364e+06     4.0  229.805921  2.915932e-172
Residual      3.995721e+06  2995.0         NaN            NaN

Significance codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1
```

Interpretation of Results

Correlation by Age and Wage

The Pearson correlation coefficient r=0.1956 indicates a weak positive linear relationship between age and wage in the dataset

The t statistic of 10.923 with 2998 degrees of freedom and an extremely small p value (~0) confirms that this correlation is statistically significant meaning we can reject the null hypothesis that there is no relationship between age and wage

The 95 percent confidence interval ranging from 0.1610 to 0.2298 tellls us that we can be 95 percent confident the true population correlation falls within this range, however the practical significance is limited because age explains only about 3.8 percent of the variance in wages calculated as r squared which equals 0.038

Normality Tests

The Shapiro Wilk test results reveal that neither variable follows a normal distribution

- Age has W statistic of 0.9911 and samll p value (~0)
- Wage has W statistic of 0.8796 and p value (~0)

Both p values are far below the 0.05 significance level leading us to reject the null hypothesis of normality for both variables

The Q-Q plots confirms these findings by showing data points deviating from the theoretical normal line especially in each ending of the distributions

Education on Wage affect

ANOVA test examining wage diferences across 5 education levels produced remarkable results

The F statistic of 229.81 is large indicating substantial diferences in mean wages among education groupes

Small p value (~0) means there is virtually no probability that these wage diferences occurred by chance alone

It all means that education level has an extremely powerful and statistically significant effect on wages making it the most important factor in this analysis

Overall

Education emerges as the dominant factor determining wages in this dataset while age plays a secondary role with limited practical impact

These findings suggest that investing in education is likely to have a much greater impact on earning potential than simply gaining work experience through age